

Assignment - 4

Course Name: Algebraic Geometry

Indian Statistical Institute, Kolkata

Note: Throughout, k denotes an algebraically closed field unless otherwise specified, and a *variety* over k (or a k -variety) is any affine, quasi-affine, projective, or quasi-projective variety.

(1) Let $P = (0, 0) \in \mathbb{A}_{\mathbb{C}}^2$. Consider the following curves in $\mathbb{A}_{\mathbb{C}}^2$ given by:

$$(i) y = x^2, \quad (ii) y^2 = x^3 + x^2, \quad (iii) y^2 = x^3.$$

For each of these curves C , do the following:

- (a) Find $\text{Sing}(C)$ —the singular locus of C .
- (b) Describe $T_{C,P}$ —the tangent cone to C at P (*Tangent cone* of a variety at a point is the union of the tangent lines at that point, counting them with their multiplicities.). Also, find the tangent lines to C at P and their multiplicities.
- (c) Find the tangent space to V at P in $\mathbb{A}^2$ (i.e. the space defined by the linear terms of the above polynomial f). Also calculate

$$\dim_{\mathbb{C}}(\mathfrak{m}_P/\mathfrak{m}_P^2)$$

directly, where $\mathfrak{m}_P$ is the maximal ideal associated to P . Compare. (Also compare to the space spanned by the tangent lines.)

- (d) Find the intersection multiplicity $(L \cdot V)_P$ of L with V at P , and find $(L \cdot V)$. [cf. Problem 17]

Does this agree with the real picture?

(2) (a) Show that

$$GL_n(k) := \{\text{invertible } n \times n \text{ matrices over } k\}$$

and

$$SL_n(k) := \{A \in GL_n(k) \mid \det(A) = 1\}$$

are group varieties over k (cf. Hartshorne, Chapter I, Exercise 3.21).

Are these varieties affine? If so, realize them explicitly as Zariski closed subsets of some affine space $\mathbb{A}_k^N$.

Is there a natural homomorphism of group varieties between them (i.e., a morphism of varieties that is also a group homomorphism)?

- (b) Consider the orthogonal group

$$O_n(k) := \{A \in GL_n(k) \mid AA^t = I\},$$

and the special orthogonal group

$$SO_n(k) := O_n(k) \cap SL_n(k).$$

Show that these are group varieties, and address the same questions as in part (a).

- (c) Discuss the unitary group in the case $k = \mathbb{C}$. To what extent can it be viewed as an algebraic group?
- (d) Compute the tangent spaces at the identity element for all the above group varieties.

Remark. Group varieties are also called *algebraic groups*. Those that are affine are called *linear algebraic groups*. Over $\mathbb{C}$, algebraic groups naturally carry the structure of complex Lie groups. In particular, $GL_n(\mathbb{C})$ and $SL_n(\mathbb{C})$ are Lie groups, and for any Lie group G , the tangent space at the identity $T_e(G)$ identifies with its Lie algebra $\mathfrak{g}$.

(3) Recall that a variety V is *rational* if it is birationally equivalent to $\mathbb{P}_k^n$ for some n (in fact, $n = \dim V$) or equivalently, if $K(V)/k$ is a purely transcendental field extension.

- (a) Prove that any conic in $\mathbb{P}_k^2$ is a rational curve.
- (b) Prove that the cuspidal cubic $\{y^2 = x^3\} \subset \mathbb{A}_k^2$ is a rational curve.

- (c) Prove that the nodal cubic curve $\{y^2 = x^3 + x^2\} \subset \mathbb{A}_k^2$ is a rational curve.
- (4) Show that the quadric surface $Q := \{xy = zw\}$ in $\mathbb{P}_k^3$ is birational to $\mathbb{P}_k^2$. Are they isomorphic?
- (5) Let X and Y be two varieties. Suppose there are points $P \in X$ and $Q \in Y$ such that the local rings $\mathcal{O}_{X,P}$ and $\mathcal{O}_{Y,Q}$ are isomorphic as k -algebras. Then show that there are open sets $P \in U \subset X$ and $Q \in V \subset Y$ and an isomorphism of U to V which sends P to Q .
- (6) Let Y be the cuspidal cubic curve $y^2 = x^3$ in $\mathbb{A}^2$. Blow up the point $O = (0, 0)$, let E be the exceptional curve, and let $\tilde{Y}$ be the strict transform of Y . Show that E meets $\tilde{Y}$ in one point, and that $\tilde{Y} \cong \mathbb{A}^1$. In this case the morphism $\varphi : \tilde{Y} \rightarrow Y$ is bijective and bicontinuous, but it is not an isomorphism.

(7) *Multiplicities of a plane curve at a point.*

Let $Y \subset \mathbb{A}^2$ be a curve defined by the equation $f(x, y) = 0$. Let $P = (a, b)$ be a point of $\mathbb{A}^2$. Make a linear change of coordinates so that P becomes the point $(0, 0)$. Then write f as a sum $f = f_0 + f_1 + \cdots + f_d$, where f_i is a homogeneous polynomial of degree i in x and y . Then we define the multiplicity of P on Y , denoted $\mu_P(Y)$, to be the least r such that $f_r \neq 0$. The linear factors of f_r are called the tangent directions at P . The following are evident:

- (i) $P \in Y \iff \mu_P(Y) > 0$.
- (ii) $\mu_P(Y) = 1 \iff P$ is a non-singular point of Y .

Intersection multiplicity

If $Y, Z \subset \mathbb{A}^2$ are two distinct curves, given by equations $f = 0$, $g = 0$, and if $P \in Y \cap Z$, we define the intersection multiplicity $(Y \cdot Z)_P$ of Y and Z at P to be the length of the $\mathcal{O}_{\mathbb{A}_k^2, P}$ -module

$$\mathcal{O}_{\mathbb{A}_k^2, P} / (f, g) \mathcal{O}_{\mathbb{A}_k^2, P}.$$

- (a) Show that $(Y \cdot Z)_P$ is finite, and $(Y \cdot Z)_P \geq \mu_P(Y) \cdot \mu_P(Z)$.
- (b) If $P \in Y$, show that for almost all lines L through P (i.e., all but a finite number), $(L \cdot Y)_P = \mu_P(Y)$.
- (c) If Y is a curve of degree d in $\mathbb{P}^2$, and if L is a line in $\mathbb{P}^2$, $L \not\subset Y$, show that $(L \cdot Y) = d$. Here we define

$$(L \cdot Y) = \sum (L \cdot Y)_P$$

taken over all points $P \in L \cap Y$, where $(L \cdot Y)_P$ is defined using a suitable affine cover of $\mathbb{P}^2$.

(8) *The affine cone over a projective variety.*

Let $Y \subset \mathbb{P}^n$ be a nonempty algebraic set, and let $\theta : \mathbb{A}^{n+1} - \{(0, \dots, 0)\} \rightarrow \mathbb{P}^n$ be the map which sends the point with affine coordinates $(a_0, \dots, a_n)$ to the point with homogeneous coordinates $(a_0 : \dots : a_n)$. We define the affine cone over Y to be

$$C(Y) = \theta^{-1}(Y) \cup \{(0, \dots, 0)\}.$$

Let $Y \subset \mathbb{P}^2$ be a nonsingular plane curve of degree > 1 , defined by the equation $f(x, y, z) = 0$. Let $X \subset \mathbb{A}^3$ be the affine variety defined by f . By the above definition, X is simply $C(Y)$. Let P be the point $(0, 0, 0)$, which is the vertex of the cone. Let $\varphi : \tilde{X} \rightarrow X$ be the blowing-up of X at P .

- (a) Show that X has just one singular point, namely P .
- (b) Show that $\tilde{X}$ is non-singular.
- (c) Show that $\varphi^{-1}(P)$ is isomorphic to Y .